SET 8: THE HOTEI REGIO CRYOVOLCANIC GEOTHERMAL TAP & HEAVY WATER EXTRACTION PLANT

DOCUMENT 8A: SECURED CREDIT AGREEMENT

SECURED TERM CREDIT AGREEMENT

Dated as of the Agreement Effective Date (as defined herein)

among

HOTEI GEOTHERMAL ENERGY & ISOTOPIC PARTNERS LP,
as Borrower

and

SOL-SYSTEM INTERPLANETARY DEVELOPMENT BANK (SSIDB),
as Administrative Agent, Collateral Agent, and Lead Arranger

and

THE FINANCIAL INSTITUTIONS PARTY HERETO FROM TIME TO TIME, as Lenders

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This SECURED TERM CREDIT AGREEMENT (this "Agreement") is entered into on the Agreement Effective Date by and among Hotei Geothermal Energy & Isotopic Partners LP, as Borrower, the Lenders party hereto from time to time, and Sol-System Interplanetary Development Bank, as Administrative Agent. In consideration of the mutual covenants and agreements herein contained, the parties hereto agree as follows:

WHEREAS, the Borrower owns and operates a comprehensive subsurface geothermal tap and cryogenic heavy water (D_2O) extraction facility situated in the cryovolcanically active Hotei Regio province of Titan; and

WHEREAS, the Borrower has requested that the Lenders make term loans to finance deep-crust borehole excavation, centrifugal isotopic separation arrays, localized seismic mitigation pylons, and active thermal transmission lines; and the Lenders are willing to do so on the terms and subject to the conditions set forth herein.

ARTICLE I: DEFINITIONS AND INTERPRETATION

Section 1.01. Defined Terms. As used in this Agreement, the following terms have the meanings specified below:

- **"Agreement Effective Date"** means the date on which this Agreement has been fully executed and delivered by all parties hereto, as verified by the date inscribed next to the signatures of the authorized officers in the execution block below (such date, the **"Signature Date"**).
- **"Cryovolcanic Tap Stability Rating"** or **"CTSR"** means the structural, thermodynamic, and pressure stability rating of the deep-bore geothermal extraction casing, as evaluated on a 100-point scale by the Titan Technical Safety Directorate (TTSD).
- **"Heavy Water Yield Target"** means the mandatory minimum operational isotopic separation metric of 17,500 kilograms of pure deuterium oxide (D_2O equivalent) per Titan sidereal day, as calculated under the provisions of Section 4.01(2) herein.
- **"Hotei Plant Substructure"** means the physical subterranean piping networks, high-pressure booster valves, centrifugal isotopic separation towers, and thermal transmission gantry arrays situated at Sector 32, Hotei Regio coordinates (latitude 26.0° S, longitude 78.0° W) on Titan, Saturn.
- **"Localized Seismic Escrow"** means the cash escrow reserve fund mandated to be maintained under Section 4.01(3) for lithospheric stress mitigation and localized structural containment.
- **"Sol System Interstellar Accounting Standards"** or **"SSIAS"** means the unified framework of financial accounting principles approved by the Interstellar Financial Board.
- **"TIDOR"** means the Titan Inter-Dome Offered Rate, which is the floating interbank benchmark rate calculated for Titan-denominated credit lines.

Section 1.02. Rules of Interpretation. Capitalized terms defined in the singular shall have the corresponding meanings in the plural, and vice versa. Any references to specific sections, tables, or documents relate directly to this Agreement unless otherwise specified.

ARTICLE II: THE SECURED CREDIT LINES

Section 2.01. The Facility. Subject to the terms and conditions of this Agreement, each Lender agrees to make a single advance term loan (each a "Facility Loan") to the Borrower on the **Closing Date** (which is identical to the **Signature Date**) in an aggregate principal amount of **8,500,000,000 Sol Credits (SC)**.

Section 2.02. Interest Rate Adjustments.

1. The Facility Loans shall accrue interest daily at a floating rate per annum equal to **TIDOR plus 7.75%**.
2. Should the **Cryovolcanic Tap Stability Rating (CTSR)** fall below 90.0% for any consecutive 3-day Titan cycle, the floating margin shall automatically increase by **3.50% per annum** (the "Thermal Runaway Containment Risk Premium") until the CTSR is restored to 93.0% or above, as verified by the TTSD.

Section 2.03. Repayment and Amortization. Repayment of the principal shall be executed in equal quarterly installments on the final operational day of each Martian fiscal quarter, starting on the first such date following the **Agreement Effective Date**, in accordance with the amortization schedules appended hereto.

ARTICLE III: REPRESENTATIONS AND WARRANTIES

The Borrower represents and warrants to the Lenders that:

Section 3.01. Subsurface Geothermal Mining Rights. The Borrower holds exclusive licenses to tap and process deep lithospheric thermal flows and extract associated chemical/isotopic compounds within Sector 32-Hotei under the authority of the Titan Regulatory Council (TRC).

Section 3.02. Cryovolcanic Pressure Tolerances. The deep geothermal extraction casings are constructed using high-density titanium-carbide alloys and active magnetic suspension, engineered to withstand external hydrostatic pressure variants up to 500 atmospheres and thermal gradients exceeding 200 Kelvin.

ARTICLE IV: COVENANTS

Section 4.01. Affirmative Covenants. So long as any part of the Facility Loan remains outstanding, the Borrower shall:

1. **Borehole Integrity Maintenance:** Undertake continuous acoustic-resonance structural audits of the cryovolcanic boreholes to ensure the CTSR does not fall below the base standard of 91.0%.
2. **Maintain Isotopic Output:** Operate the centrifugal separating columns to sustain the **Heavy Water Yield Target** of 17,500 kilograms per Titan day, ensuring supply covenants with orbital transport transport vessels are satisfied.
3. **Localized Seismic Escrow Account:** Maintain the cash-funded **Localized Seismic Escrow** with the Administrative Agent, holding a balance of not less than SC 500,000,000 at all times to manage ocean-floor structural anchoring stability.

Section 4.02. Negative Covenants. The Borrower shall not, without the explicit written approval of the Administrative Agent:

1. **System Modifications:** Reroute or depressurize the containment casings of any active secondary geothermal taps within the designated coordinates.
2. **Restricted Payments:** Declare or make any dividend distributions to corporate parent entities if the CTSR is below 92.0% or if a Default Event has occurred and is continuing.

ARTICLE V: DEFAULT EVENTS AND REMEDIES

Section 5.01. Default Events. The occurrence of any of the following events shall constitute an Event of Default:

1. **Thermal Borehole Blowout:** Any sudden structural casing failure leading to an uncontrollable high-pressure cryovolcanic blowout exceeding 25,000 metric tons of ammonia-rich ice slurry within any 24-hour cycle.
2. **Covenant Non-Compliance:** Failure to maintain the required balance in the **Localized Seismic Escrow** under Section 4.01(3), or failure to meet the **Heavy Water Yield Target** for more than two consecutive quarters.
3. **Bedrock Dissolution:** Any localized sub-surface heating adjacent to the primary borehole casing causing structural water-ice bedrock temperatures to exceed 245 Kelvin, threatening the physical structural integrity of the plant foundation.

ARTICLE VI: MISCELLANEOUS PROVISIONS

Section 6.01. Governing Law. This Agreement shall be governed by, and construed in accordance with, the Admiralty Law of the Martian Lowlands and the outer-rim industrial treaties governing the Saturnian System.

Section 6.02. Counterparts. This Agreement may be executed in any number of counterparts, each of which when so executed and delivered shall be deemed an original, but all of which together shall constitute one and the same instrument.

HOTEI GEOTHERMAL & ISOTOPIC PARTNERS LP

By: _____
Name: Julian Kase-Vance
Title: Managing Partner, Geothermal Division

Date of Execution: **June 26, 2146**

SOL-SYSTEM INTERPLANETARY DEVELOPMENT BANK

By: _____
Name: Diana Sterling-Lowell
Title: Managing Director, Outer-Rim Assets Division

Date of Execution (the **Signature Date**): **June 26, 2146**